Week 14 — 🎉 Big Simulations

Name: ____ Date: ____

> ***Monte Carlo method:*** *answer a hard question with a mountain of random tries. Fraction of raindrops inside = fraction of area inside. Rain can measure.*

Today's words

Word	What it means
Monte Carlo method	answering a hard question with tons of random tries
`random.uniform(a, b)`	a random *decimal* between a and b — any decimal at all
trial	one single random try (one raindrop, one game)
expected value	your average result *per game*: each outcome × its probability, added up

1 · Be the raindrop 🌧️

The quarter circle's arc runs from (0, 1) to (1, 0) — every arc point is exactly 1 from the corner (0, 0). Sketch the arc on the grid, plot the five drops, then **compute**: inside means $x^2 + y^2 \le 1$.

```
  y
1.0 ┌──────────────┐
    │              │
0.8 │              │
    │              │
0.6 │              │
    │              │
0.4 │              │
    │              │
0.2 │              │
    │              │
0.0 └──────────────┘
   0.0  0.2  0.4  0.6  0.8  1.0   x
```

Drop	x	y	x^2	y^2	$x^2 + y^2$	≤ 1? in or out?
A	0.3	0.4				
B	0.6	0.8				
C	0.9	0.9				
D	0.5	0.5				
E	0.7	0.8				

2 · Audit the carnival 🎪

Lucky Die: pay **$4**, roll one die, win $1 per dot. Fill in the table, then the verdict.

Roll	Payout	Probability	Payout × probability
1	$1	1/6	$
2	$2	1/6	$
3	$3	1/6	$
4	$4	1/6	$
5	$5	1/6	$
6	$6	1/6	$

Expected payout (add the last column): **$__**

Per game at $4 a ticket, a player averages: **$__** · After 1,000 games, the booth is up about: **$__**

🧠 Brain teaser (optional — take it home)

Design a sneaky carnival game. Invent a game (coins, dice, spinners — your call) that *looks* totally fair but has a **negative expected value** for the player. Write:

1. The rules and the ticket price (make the sign tempting!)
2. The expected-value math that exposes the trick

Bring it next week — **the sneakiest game gets simulated in class**, 100,000 plays, so the numbers better back you up.